User Stories

We plan our work in small, customer-centric pieces.

Our Journey Today

The Foundation
What is a user story and why does it exist?

The Craft
How do we write a high-quality story?

The Management
How do we work with stories in a project?

The Evolution
How does a story drive development and testing?

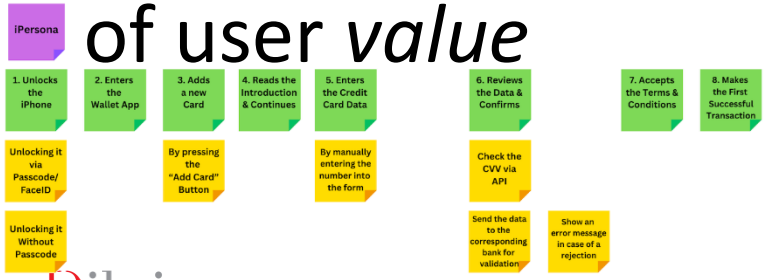
Next Steps
Your Flipped Classroom Assignment

PART 1: THE FOUNDATIONS

Where We Came From: The Limits of *The Spec*



- Massive, detailed specification documents
- Freeze knowledge at the *beginning* of the development
- Often written *in isolation* from the development team
- Rigid, difficult to change, and quickly outdated
- Focus on *features* and *functions*, often losing sight of user *value*



Dibris

A New Approach:
A Placeholder for Conversation

- A user story is a short, simple description of a feature told from the perspective of the person who desires it
  - it describes a *need*
  - it is **not** a detailed specification
- Its primary purpose is to articulate value and initiate a conversation

A user story is an invitation to a conversation
Alistair Cockburn

The Anatomy of a Story

- As a <role>
- I want <goal>
- So that <benefit>

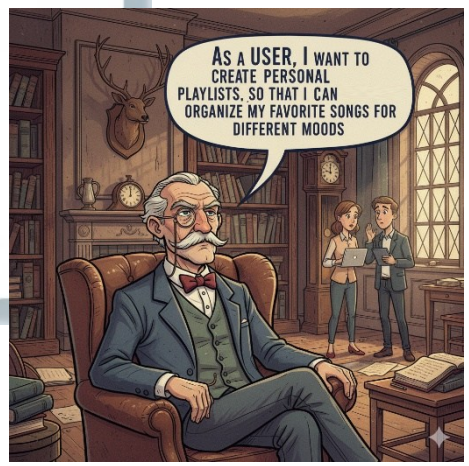
Who are we building this for? (The User)

What are they trying to achieve? (The Action)

Why do they want it? (The Value)

User Story

As a music lover, I want to create personal playslists, so that I can I can organize my favorite songs for different moods.

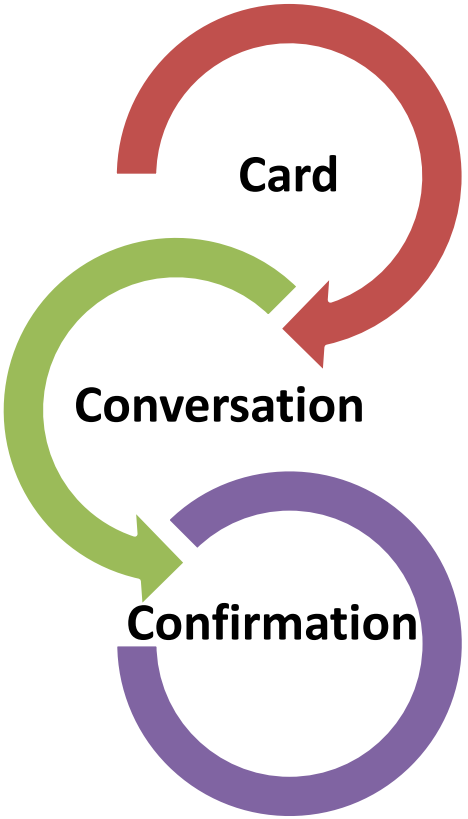




PART 2: THE CRAFT

The Lifecycle of a Story: Card, Conversation, Confirmation

- **Card**: A physical (or digital) token
  - the essence of the requirement
- **Conversation**: A *verbal* exchange between the team, product owner, and stakeholders to flesh out the details
  - building shared understanding
- **Confirmation**: An agreement on how to verify the story is "done" correctly
  - **Acceptance Criteria**



Usually completed over time through many sessions

How We Know We're Done

- Acceptance Criteria (AC) = a set of testable conditions that must be met for the story to be considered complete
  - They transform ambiguity into a clear, shared checklist
  - Written on the back of the card
- **Acceptance Criteria**

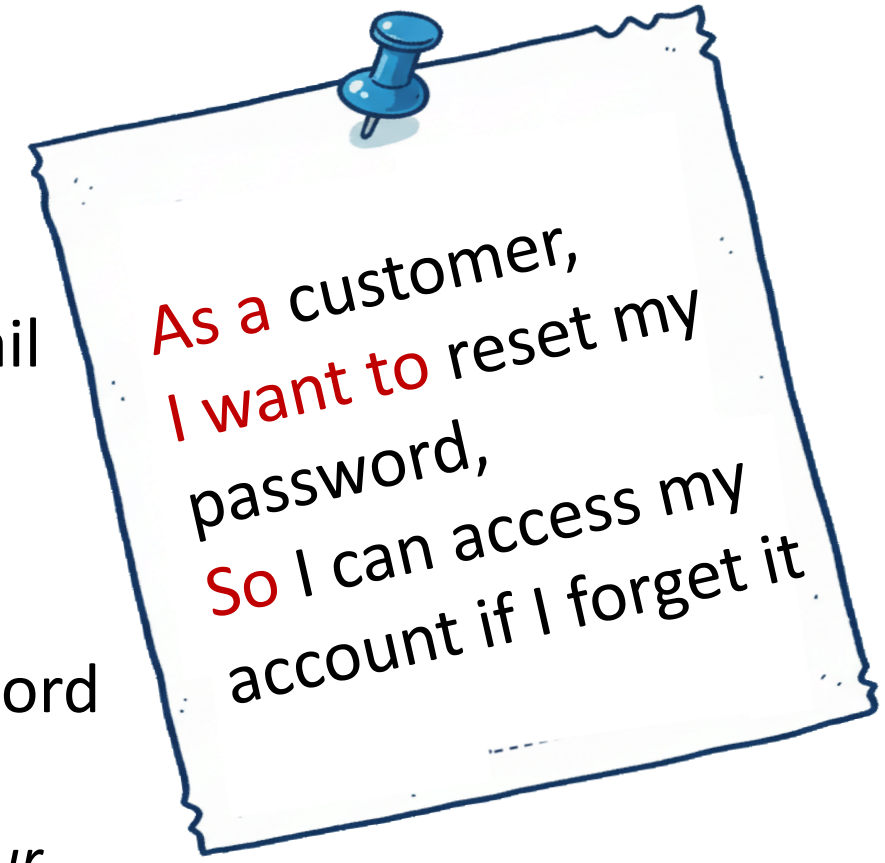
Given I am on the login page,

When I click "Forgot Password,"
Then I am prompted for my email

When I submit a valid email,
Then a confirmation message is shown

Then a unique, single-use password reset link is sent to my email

The link must expire in 1 hour



A Checklist for a Good Story

- **I – Independent** Can be developed and delivered on its own
- **N - Negotiable** Not a rigid contract; details can be discussed
- **V - Valuable** Delivers tangible value to the end-user
- **E - Estimable** The team can give a rough size estimate
- **S - Small** Can be completed within a single development iteration
- **T - Testable** You can define clear acceptance criteria for it

From Big Ideas to Manageable Work

- **Epic**: A very large story that represents a major feature. It's too big to complete in one sprint
- **Story**: A smaller piece of functionality broken out from an Epic that fits the INVEST criteria
- **Task**: A specific technical action needed to complete a story




coffee

THE MANAGEMENT

Sizing the Work: The Problem with Time

- **Why not estimate in hours?**
  - Humans are poor at absolute estimation
  - An hour for a senior developer is different from an hour for a junior
  - Time estimates can be perceived as commitments, creating pressure
- **Enter Story Points**
  - A relative measure of effort, complexity, and uncertainty
  - Often uses a modified Fibonacci sequence: 1, 2, 3, 5, 8, 13...
    - sometimes visualized graphically as T-Shirt sizes, country sizes, buckets…
  - A “2” is roughly twice the effort of a “1”, an “8” is a big, complex story
  - A reference story is needed to establish the effort of a “1”

Collaborative Estimation

A consensus-based technique to estimate stories

- **Discuss**
  - Explanations by the Product Owner
  - Q&A with the team
- **Estimate**
  - Each person *privately* chooses a card with a story point value
- **Reveal**
  - All cards are shown at the same time



- **Align**
  - If estimates differ, the highest and lowest are explained
  - To share knowledge
- **Repeat**
  - until a consensus is reached

Deciding What to Build Next

- A simple but powerful framework for prioritizing the product backlog.
- Plot stories on a 2x2 matrix.

Value		
High	Quick Wins	Big Bets
Low	Fill-ins	Money Pits Reconsider
	Low	High **Effort**

PART 4: THE EVOLUTION

The Story's Final Form: Executable Specifications

- What if our Acceptance Criteria could be more than just a checklist?
- What if they could be directly transformed into automated tests?
- This is the core idea behind **Behavior-Driven Development (BDD)**

A Common Language for Business and Tech

- BDD uses a structured, human-readable language called **Gherkin**

  ⇒ understandable by product owners/users

- It follows a Given / When / Then format

  Given: The initial context (the state of the system)

  When: The action the user performs

  Then: The expected outcome or result

  – same structure as a test

  ⇒ test-as-specification for developers

  ⇒ easy to [automatically] translate into a test

BDD Example

Card Title: Add item to cart from search

Card Front

As a shopper,

I want to add an item to my cart from the search results page,

so that I can continue shopping without visiting the product page

Card Back

Given a user is on the search results page for "agile books“

And the "User Stories Applied" book is visible

When the user clicks the "Add to Cart" button for that book

Then the cart icon in the header should display "1 item"

CONCLUSION

The Power of a Story

- **User Stories' Strong Points**
  - **Focus on Value**: They force us to ask "why?"
  - **Foster Collaboration**: They are invitations to a conversation
  - **Enable Iteration**: They are small, manageable units of work
  - **Drive Quality**: They connect directly to testable acceptance criteria

The Flipped Classroom

- On Aulaweb you will find
  - These slides
  - The list of resources everybody **MUST** read.
  - The calendar for next lesson
  - A choice to select your topic
- Each lesson will address one area
  - The volunteers for (topics in) that area will research it further, present it in a 10-minute talk, and lead the discussion
  - After the discussion we will do some hands-on exercise

Focus points

- The Foundations
  - Definition of user story
  - Their role in agile development
  - User roles/personas
  - Good/bad examples



- The craft
  - Lifecycle (e.g., C^3)
  - Acceptance criteria
  - Size



- The Management (why, when, why, how)
  - Estimation
    - Story points
  - Prioritization



- The Evolution
  - Gherkin
  - Test automation
  - BDD



Chose your Area, the volunteers will coordinate to distribute contents